

ZEWAIL CITY OF SCIENCE & TECHNOLOGY

DEPARTMENT OF AEROSPACE ENGINEERING

SPC 337: GAS DYNAMICS

Design of Ramjet Engine

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Abstract

This project presents the design and optimization of a Ramjet engine operating at a flight Mach number of 2.75 at an altitude corresponding to 12.11 kPa ambient pressure. The primary objective was to maximize net thrust by optimizing the intake geometry (two-ramp supersonic inlet) and the nozzle area ratio under a fixed square cross-section constraint (25×25 cm). A MATLAB-based grid search algorithm was implemented to solve the coupled 2D oblique shock and 1D flow equations. The optimized design features intake ramp angles of 14.50° and 17.50° , achieving a total pressure recovery of 0.8086. The engine produces a net thrust of 6.26 kN with a specific thrust of 633.32 N/(kg/s) at a maximum combustor temperature of 2200 K. Off-design analysis at Mach 2.0 and Mach 4.0 demonstrates the operational limits and performance variations of the fixed-geometry engine.

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1 Introduction

The Ramjet is a form of air-breathing jet engine that uses the engine's forward motion to compress incoming air without a rotary compressor. This project involves the preliminary design of a Ramjet engine sized for a 0.0625 m^2 square cross-section. The design process integrates the analysis of a supersonic intake, subsonic diffuser, Rayleigh flow combustor, and a converging-diverging nozzle.

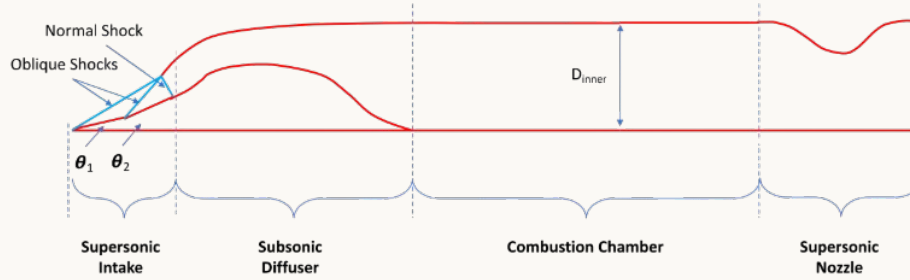


Figure 1: Schematic of the Ramjet engine stations.

1.1 Objectives

- Design a 2-ramp supersonic intake to maximize total pressure recovery.
- Optimize the nozzle area ratio for maximum thrust within geometric constraints.
- Analyze the engine performance at the design point ($M_\infty = 2.75$).
- Evaluate off-design performance at $M_\infty = 2.0$ and 4.0 for fixed geometry.

1.2 Assumptions

Model Assumptions

- **Gas Properties:** Air is treated as a calorically perfect gas ($\gamma = 1.4$, $R = 287 \text{ J/kg}\cdot\text{K}$).
- **Flow Dimensionality:** 2D flow for inlet ramps; quasi-1D flow for diffuser, combustor, and nozzle.
- **Combustor:** Modeled as 1D Rayleigh flow with an additional 5% total pressure loss due to friction.
- **Adiabatic Flow:** The intake, diffuser, and nozzle are assumed adiabatic.
- **Geometry:** The engine has a square cross-section (Side = 25 cm).

2 Methodology

The design was performed using a custom MATLAB script utilizing a grid-search optimization algorithm.

2.1 Governing Relations and Station Definitions

The engine flowpath is discretized into the stations shown in Fig. 1: freestream (0), post-oblique shocks (1–2), post-terminal normal shock (3), diffuser exit (4), combustor exit (5), nozzle throat (6), and nozzle exit (7). Air is modeled as a calorically perfect gas with constant γ and R .

2.1.1 Isentropic Relations

For any quasi-1D isentropic segment, the total (stagnation) properties are

$$\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2, \quad (1)$$

$$\frac{P_t}{P} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}}, \quad (2)$$

and the speed of sound and velocity are

$$a = \sqrt{\gamma R T}, \quad V = Ma. \quad (3)$$

2.1.2 Captured Mass Flow at Design Point

Assuming full capture (no spillage) at the inlet plane,

$$\dot{m} = \rho_\infty V_\infty A_{cap} = \frac{P_\infty}{RT_\infty} M_\infty \sqrt{\gamma R T_\infty} A_{cap}. \quad (4)$$

2.2 Two-Ramp Supersonic Intake (Oblique–Oblique–Normal)

Each ramp produces an attached weak oblique shock, followed by a terminal normal shock. For a given upstream Mach number M_1 and flow deflection θ , the shock angle β is obtained from the θ – β – M relation:

$$\tan \theta = 2 \cot \beta \frac{M_1^2 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos 2\beta) + 2}. \quad (5)$$

The normal component is $M_{n1} = M_1 \sin \beta$ and the downstream normal Mach number follows the normal-shock relation

$$M_{n2}^2 = \frac{1 + \frac{\gamma - 1}{2} M_{n1}^2}{\gamma M_{n1}^2 - \frac{\gamma - 1}{2}}, \quad (6)$$

with the downstream Mach number

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)}. \quad (7)$$

Static pressure rise across each oblique shock is computed using the corresponding normal-shock expression:

$$\frac{P_2}{P_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n1}^2 - 1). \quad (8)$$

2.2.1 Total Pressure Loss Across Shocks

Shock irreversibility is quantified by the total pressure ratio across a normal shock (evaluated using M_{n1}):

$$\frac{P_{t2}}{P_{t1}} = \left(\frac{(\gamma + 1) M_{n1}^2}{2 + (\gamma - 1) M_{n1}^2} \right)^{\frac{\gamma}{\gamma - 1}} \left(\frac{\gamma + 1}{2\gamma M_{n1}^2 - (\gamma - 1)} \right)^{\frac{1}{\gamma - 1}}. \quad (9)$$

The intake total pressure recovery is therefore

$$\Pi_{\text{rec}} \equiv \frac{P_{t,3}}{P_{t,0}} = \left(\frac{P_{t,1}}{P_{t,0}} \right) \left(\frac{P_{t,2}}{P_{t,1}} \right) \left(\frac{P_{t,3}}{P_{t,2}} \right). \quad (10)$$

2.3 Terminal Normal Shock and Subsonic Diffuser

The terminal normal shock processes the flow from station 2 to 3. For upstream Mach M_2 , the downstream Mach number is

$$M_3^2 = \frac{1 + \frac{\gamma-1}{2}M_2^2}{\gamma M_2^2 - \frac{\gamma-1}{2}}, \quad (11)$$

and the corresponding pressure ratio is given by Eq. (8) with $M_{n1} \rightarrow M_2$.

The subsonic diffuser (3→4) is assumed adiabatic and (ideally) isentropic, hence

$$T_{t,4} = T_{t,3}, \quad P_{t,4} \simeq P_{t,3}, \quad (12)$$

and p_4 , T_4 follow from Eqs. (1)–(2) for the chosen diffuser exit Mach number M_4 .

2.4 Nozzle Sizing and Thrust

The nozzle is modeled as quasi-1D, adiabatic, and isentropic. The throat is assumed choked ($M_6 = 1$). The choked mass-flow relation gives the throat area:

$$\dot{m} = P_{t,5} A_t \sqrt{\frac{\gamma}{R T_{t,5}}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \Rightarrow A_t = \frac{\dot{m}}{P_{t,5}} \sqrt{\frac{R T_{t,5}}{\gamma}} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}. \quad (13)$$

For a given exit Mach number M_e , the area ratio is

$$\frac{A_e}{A_t} = \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}. \quad (14)$$

In the ideal case, the nozzle is designed for perfect expansion ($p_e = p_a$) at the design point; however, the square cross-section imposes the geometric constraint $A_e \leq A_{\max}$.

The nozzle exit velocity is

$$V_e = M_e \sqrt{\gamma R T_e}, \quad T_e = \frac{T_{t,5}}{1 + \frac{\gamma-1}{2} M_e^2}. \quad (15)$$

2.4.1 Net Thrust

The net thrust is computed from a control-volume momentum balance:

$$F_{\text{net}} = \dot{m}(V_e - V_\infty) + (p_e - p_a)A_e \quad (16)$$

and the specific thrust is $F_{\text{net}}/\dot{m}$.

2.5 Combustor Modeling via Rayleigh Flow

Heat addition in the constant-area combustor is modeled as 1D Rayleigh flow (station 4→5). A key feature is the choking (Rayleigh) limit: for subsonic inlet flow ($M_4 < 1$), heat addition increases Mach number toward unity.

Define the Rayleigh starred condition (*) as the state where $M = 1$. A convenient nondimensional

Rayleigh relation for stagnation temperature is

$$\frac{T_t}{T_t^*} = \frac{2(\gamma + 1)M^2 \left(1 + \frac{\gamma-1}{2}M^2\right)}{(1 + \gamma M^2)^2}. \quad (17)$$

Given M_4 and $T_{t,4}$, Eq. (17) determines T_t^* and thus the maximum allowable combustor exit total temperature

$$T_{t,5} \leq \min(T_{t,\max}, T_t^*), \quad (18)$$

where $T_{t,\max} = 2200$ K is the imposed material/combustion constraint.

Total pressure loss in Rayleigh flow is captured by the standard Rayleigh P_t/P_t^* function, and an additional empirical friction loss is imposed:

$$P_{t,5} = 0.95 P_{t,5,\text{Rayleigh}}. \quad (19)$$

Rayleigh heat addition produces a substantial entropy rise, consistent with

$$\Delta s_{4 \rightarrow 5} = c_p \ln \left(\frac{T_5}{T_4} \right) - R \ln \left(\frac{p_5}{p_4} \right) > 0. \quad (20)$$

2.6 Optimization Formulation

The design problem is posed as a constrained optimization:

$$\max_{\theta_1, \theta_2, A_e/A_t, T_{t5}} F_{\text{net}}(\theta_1, \theta_2, A_e/A_t, T_{t5}) \quad (21)$$

$$\text{s.t. } \theta_i \leq \theta_{\max}(M, \gamma) \text{ (attached weak-shock condition),} \quad (22)$$

$$T_{t,5} \leq 2200 \text{ K,} \quad (23)$$

$$A_e \leq A_{\max} = 0.25^2 \text{ m}^2, \quad (24)$$

$$A_t \leq A_{\max}, \quad (25)$$

$$M_6 = 1 \text{ (choked throat).} \quad (26)$$

A grid search over (θ_1, θ_2) was employed to ensure robustness and to avoid convergence to nonphysical roots.

3 Results and Discussion

3.1 Optimized Geometry

The optimization yielded the following geometric parameters. The intake ramp angles were selected to balance shock strength and total pressure recovery.

Table 1: Optimized Geometrical Dimensions.

Parameter	Value	Unit
Ramp Angle 1 (θ_1)	14.50	deg
Ramp Angle 2 (θ_2)	17.50	deg
Capture Area (A_{cap})	0.0625	m ²
Nozzle Throat Area (A_t)	0.0562	m ²
Nozzle Exit Area (A_e)	0.0625	m ²
Wedge 1 Length	0.2910	m
Wedge 2 Length	0.1204	m

3.2 Performance Metrics

At the design condition ($M_\infty = 2.75$), the engine performance is summarized below. The intake pressure recovery of 0.8086 indicates efficient compression.

Table 2: Engine Performance at Design Point.

Metric	Value
Intake Pressure Recovery (P_{t3}/P_{t0})	0.8086
Combustor Total Temperature (T_{t5})	2200.0 K
Mass Flow Rate ($\dot{m}$)	9.88 kg/s
Net Thrust (F_{net})	6257.13 N
Specific Thrust	633.32 N/(kg/s)

3.3 Station Properties

Detailed flow properties at every major station are presented in Table 3. The temperature rise across the shocks is physically consistent, ensuring valid density and area calculations.

Table 3: Flow Properties at Major Stations.

Station	Mach	P [kPa]	T [K]	Pt [kPa]	Tt [K]
0 (Freestream)	2.75	12.11	216.5	304.45	544.0
1 (After Oblique 1)	2.09	31.02	290.1	280.17	544.0
2 (After Oblique 2)	1.42	78.64	387.2	258.53	544.0
3 (After Normal Shock)	0.73	172.68	491.5	246.19	544.0
4 (Diffuser Exit)	0.22	238.02	538.7	246.19	544.0
5 (Combustor Exit)	0.61	158.31	2046.5	203.92	2200.0
6 (Nozzle Throat)	1.00	107.73	1833.3	203.92	2200.0
7 (Nozzle Exit)	1.39	64.63	1584.3	203.92	2200.0

3.4 Thermodynamic Visualization on the T - s Plane

The T - s diagram in Fig. 2 is constructed using the perfect-gas entropy relation (relative to station 0):

$$s_i - s_0 = c_p \ln \left(\frac{T_i}{T_0} \right) - R \ln \left(\frac{p_i}{p_0} \right) \quad (27)$$

highlighting the entropy increase across shocks and within the combustor.

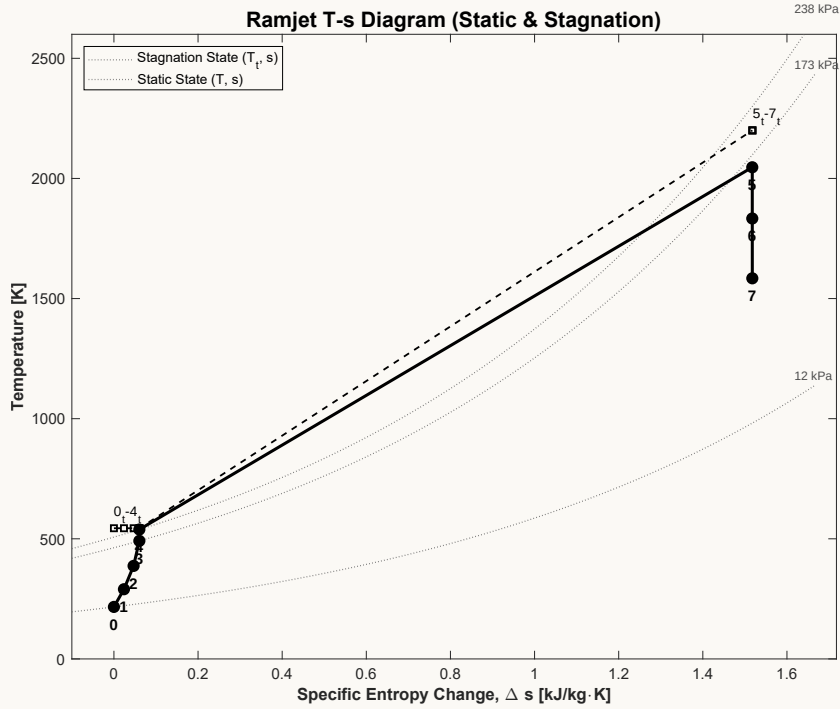


Figure 2: Temperature–Entropy diagram for the designed Ramjet cycle.

4 Bonus: Off-Design Analysis (Fixed Geometry)

Off-design performance was evaluated at $M_\infty = 2.0$ and $M_\infty = 4.0$ while keeping the inlet and nozzle geometry fixed at the design values ($\theta_1, \theta_2, A_t, A_e$). For each off-design Mach number, the inlet shock system is recomputed, followed by diffuser and Rayleigh combustor calculations. The combustor exit total temperature $T_{t,5}$ is treated as the control variable (fuel flow), and is chosen to maximize net thrust subject to $T_{t,5} \leq 2200$ K.

4.1 Criterion for Intake Start/Unstart

In the present model, each inlet ramp is assumed to generate an *attached* weak oblique shock. For a given upstream Mach number M and flow deflection θ , the shock angle β satisfies the θ – β – M relation

$$\tan \theta = 2 \cot \beta \frac{M^2 \sin^2 \beta - 1}{M^2(\gamma + \cos 2\beta) + 2}. \quad (28)$$

An attached solution exists only if $\theta \leq \theta_{\max}(M, \gamma)$, where $\theta_{\max}$ is the maximum deflection angle attainable by the weak-shock branch. Thus, the *start condition* for ramp i is

$$\theta_i \leq \theta_{\max}(M_{i-1}, \gamma), \quad i \in \{1, 2\}, \quad (29)$$

with M_{i-1} being the Mach number upstream of ramp i (note that M_1 is reduced from M_∞ due to the first ramp).

4.2 Mach 2.0 Case: Shock-2 Detachment (Unstart)

At $M_\infty = 2.0$, ramp 1 remains attached; however, the downstream Mach number after the first shock is reduced sufficiently that the second ramp deflection exceeds the allowable attached limit, i.e.,

$$\theta_2 > \theta_{\max}(M_1, \gamma), \quad (30)$$

and no weak attached solution to Eq. (28) exists. In practice, this corresponds to a detached shock (bow shock) on the second ramp, which invalidates the designed shock train and is classified as *intake unstart* in this simplified model.

```

1  --- MACH 2.0 FAILURE ANALYSIS ---
2  M_inf: 2.00
3  Theta 1: 14.50 deg -> M1: 1.466
4  Design Theta 2: 17.50 deg
5  Max Theta possible at M1: 11.22 deg
6  RESULT: UNSTART (Theta 2 > Theta Max)

```

4.3 Mach 4.0 Case: Optimal Combustor Temperature

For $M_\infty = 4.0$, the fixed-geometry inlet remains started and the combustor temperature $T_{t,5}$ is swept to maximize net thrust:

$$T_{t,5,\text{opt}} = \arg \max_{T_{t,5} \leq 2200 \text{ K}} F_{\text{net}}(T_{t,5}). \quad (31)$$

The net thrust is computed from the control-volume momentum balance

$$F_{\text{net}} = \dot{m}(V_e - V_\infty) + (p_e - p_a)A_e, \quad (32)$$

where V_e depends on $T_{t,5}$ through the nozzle isentropic relation

$$V_e = M_e \sqrt{\gamma R T_e}, \quad T_e = \frac{T_{t,5}}{1 + \frac{\gamma-1}{2} M_e^2}. \quad (33)$$

With fixed nozzle geometry (A_e/A_t), the exit Mach number M_e is determined from the area–Mach relation, and the dominant effect of increasing $T_{t,5}$ is an increase in $V_e \propto \sqrt{T_{t,5}}$. In the present design, this monotonic increase persists up to the temperature constraint, yielding

$$T_{t,5,\text{opt}} = 2200 \text{ K} \quad (34)$$

and a maximum net thrust of $F_{\text{net,max}} = 13.67 \text{ kN}$.

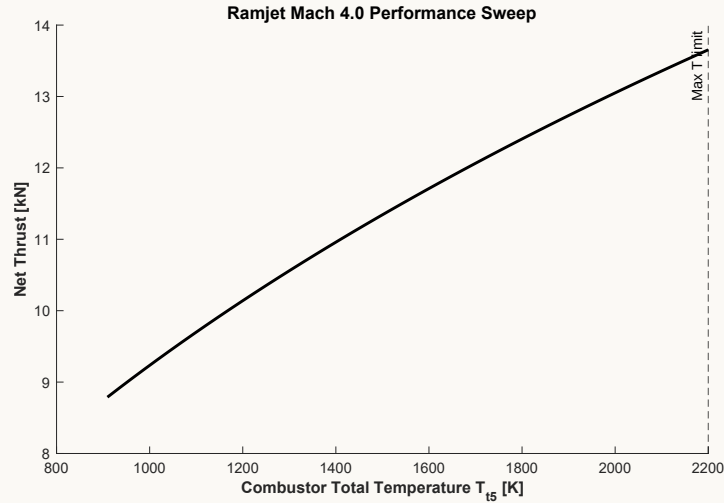


Figure 3: Optimum lies at upper bound.

5 Conclusion

The design of a Ramjet engine for Mach 2.75 flight was successfully optimized. A two-ramp intake with angles 14.5° and 17.5° provided a robust pressure recovery of 81%. The engine delivers approximately 6.26 kN of thrust within the specified geometric envelope. Off-design analysis highlights the limitations of fixed-geometry inlets at lower Mach numbers (unstart at Mach 2.0) while demonstrating significant thrust potential at higher speeds (Mach 4.0).

Appendix A: MATLAB Code

Listing 1: Ramjet Optimization Script

```

1 clear; clc; close all;
2 gamma = 1.4;
3 R = 287;
4 Cp = 1004;
5 M_inf = 2.75;
6 P_inf = 12.11e3;
7 T_inf = 216.5;
8 Side_Length = 0.25;
9 A_cross_max = Side_Length^2;
10 T_comb_max = 2200;
11 theta_range = 3:0.5:22;
12 fprintf('Running Design Optimization (Grid Search)...\n');
13 best_F_net = -inf;
14 best_design = [];
15 valid_count = 0;
16 for theta1 = theta_range
17     for theta2 = theta_range
18
19         Tt0 = T_inf * (1 + (gamma-1)/2 * M_inf^2);
20         Pt0 = P_inf * (1 + (gamma-1)/2 * M_inf^2)^(gamma/(gamma-1));

```

```

21     A0 = A_cross_max;
22
23     rho_inf = P_inf / (R * T_inf);
24     V_inf = M_inf * sqrt(gamma * R * T_inf);
25     mdot = rho_inf * V_inf * A0;
26
27     [~, M1, pt1_rat, valid1] = solve_oblique_robust(M_inf, theta1, gamma);
28     if ~valid1, continue; end
29
30     Pt1 = Pt0 * pt1_rat;
31     Tt1 = Tt0;
32     T1 = Tt1 / (1 + (gamma-1)/2 * M1^2);
33     P1 = Pt1 / ((1 + (gamma-1)/2 * M1^2)^(gamma/(gamma-1)));
34
35     [~, M2, pt2_rat, valid2] = solve_oblique_robust(M1, theta2, gamma);
36     if ~valid2, continue; end
37
38     Pt2 = Pt1 * pt2_rat;
39     Tt2 = Tt0;
40     T2 = Tt2 / (1 + (gamma-1)/2 * M2^2);
41     P2 = Pt2 / ((1 + (gamma-1)/2 * M2^2)^(gamma/(gamma-1)));
42
43     [M3, pt3_rat] = solve_normal(M2, gamma);
44
45     Pt3 = Pt2 * pt3_rat;
46     Tt3 = Tt0;
47     T3 = Tt3 / (1 + (gamma-1)/2 * M3^2);
48     P3 = Pt3 / ((1 + (gamma-1)/2 * M3^2)^(gamma/(gamma-1)));
49
50     Intake_Recovery = Pt3 / Pt0;
51
52     M4 = 0.22;
53
54     Pt4 = Pt3;
55     Tt4 = Tt3;
56     P4 = Pt4 / ((1 + (gamma-1)/2 * M4^2)^(gamma/(gamma-1)));
57     T4 = Tt4 / (1 + (gamma-1)/2 * M4^2);
58
59
60     ray_rat_4 = (2*(gamma+1)*M4^2 * (1 + (gamma-1)/2*M4^2)) / (1 + gamma*M4^2)^2;
61     Tt_star = Tt4 / ray_rat_4;
62
63     if T_comb_max > Tt_star
64         Tt5 = Tt_star;
65         M5 = 1.0;
66     else
67         Tt5 = T_comb_max;
68         ray_rat_5 = Tt5 / Tt_star;
69         M5 = solve_rayleigh_M(ray_rat_5, gamma);
70     end
71
72     pt_star_rat_4 = (gamma+1)/(1+gamma*M4^2) * ((2/(gamma+1)) * (1+(gamma-1)/2*M4^2))^(gamma/(gamma-1));

```

```

73     pt_star_rat_5 = (gamma+1)/(1+gamma*M5^2) * ((2/(gamma+1)) * (1+(gamma-1)/2*M5
74         ^2))^(gamma/(gamma-1));
75
76     Pt5_rayleigh = Pt4 * (pt_star_rat_5 / pt_star_rat_4);
77
78     Pt5 = 0.95 * Pt5_rayleigh;
79
80     sonic_const = sqrt(gamma/R) * (2/(gamma+1))^(((gamma+1)/(2*(gamma-1))));
81     A_throat = mdot / (Pt5 * sonic_const / sqrt(Tt5));
82
83     if A_throat > A_cross_max
84         continue;
85     end
86
87     Pa = P_inf;
88     if Pa >= Pt5
89         F_net = -inf;
90     else
91         Me_ideal = sqrt( ( (Pt5/Pa)^((gamma-1)/gamma) - 1 ) * 2/(gamma-1) );
92
93         Ae_At_ideal = (1/Me_ideal) * ((2/(gamma+1)) * (1 + (gamma-1)/2*Me_ideal^2)
94             ^((gamma+1)/(2*(gamma-1))));
95         Ae_required = Ae_At_ideal * A_throat;
96
97         if Ae_required > A_cross_max
98             Ae = A_cross_max;
99             Ae_At = Ae / A_throat;
100             [Me, ~] = solve_area_mach(Ae_At, gamma, 'supersonic');
101             Pe = Pt5 * (1 + (gamma-1)/2*Me^2)^(-gamma/(gamma-1));
102         else
103             Ae = Ae_required;
104             Me = Me_ideal;
105             Pe = Pa;
106         end
107
108         Ve = Me * sqrt(gamma * R * (Tt5 / (1 + (gamma-1)/2*Me^2)));
109
110         F_gross_term = mdot * Ve + (Pe - Pa) * Ae;
111
112         F_net = F_gross_term - (mdot * V_inf);
113
114         valid_count = valid_count + 1;
115     end
116
117     if F_net > best_F_net
118         best_F_net = F_net;
119
120         best_design.theta1 = theta1;
121         best_design.theta2 = theta2;
122         best_design.recovery = Intake_Recovery;
123         best_design.F_net = F_net;
124         best_design.mdot = mdot;
125         best_design.Ae = Ae;

```

```

124     best_design.At = A_throat;
125     best_design.Tt5 = Tt5;
126
127     Te = Tt5 / (1 + (gamma-1)/2*Me^2);
128
129     best_design.M = [M_inf, M1, M2, M3, M4, M5, 1.0, Me];
130     best_design.Pt = [Pt0, Pt1, Pt2, Pt3, Pt4, Pt5, Pt5, Pt5];
131     best_design.Tt = [Tt0, Tt0, Tt0, Tt0, Tt4, Tt5, Tt5, Tt5];
132     best_design.P = [P_inf, P1, P2, P3, P4, Pt5/((1+(gamma-1)/2*M5^2)^(gamma/(
        gamma-1))), Pt5*(2/(gamma+1))^(gamma/(gamma-1)), Pe];
133     best_design.T = [T_inf, T1, T2, T3, T4, Tt5/(1+(gamma-1)/2*M5^2), Tt5*(2/(
        gamma+1)), Te];
134
135     rho2 = P2 / (R * T2);
136     V2 = M2 * sqrt(gamma * R * T2);
137     A2 = mdot / (rho2 * V2);
138
139     h_cap = sqrt(A0);
140     h_2 = A2 / h_cap;
141     dh = h_cap - h_2;
142
143     best_design.L_wedge1 = (0.5 * dh) / tand(theta1);
144     best_design.L_wedge2 = (0.5 * dh) / tand(theta1 + theta2);
145     end
146 end
147 end
148 fprintf('\n=====\\n');
149 fprintf('OPTIMIZED_RAMJET_DESIGN_REPORT\\n');
150 fprintf('=====\\n');
151 if isempty(best_design)
152     fprintf('ERROR: No valid design found.\\n');
153 else
154     fprintf('Optimized Intake Angles:\\n');
155     fprintf('Theta1: %.2f deg\\n', best_design.theta1);
156     fprintf('Theta2: %.2f deg\\n', best_design.theta2);
157     fprintf('\\nPerformance Metrics:\\n');
158     fprintf('Intake Pressure Recovery: %.4f\\n', best_design.recovery);
159     fprintf('Combustor Total Temp: %.1f K\\n', best_design.Tt5);
160     fprintf('Net Thrust: %.2f N\\n', best_design.F_net);
161     fprintf('Specific Thrust: %.2f N/(kg/s)\\n', best_design.F_net/best_design.mdot)
        ;
162
163     fprintf('\\nGeometry:\\n');
164     fprintf('Capture Area: %.4f m^2\\n', A_cross_max);
165     fprintf('Nozzle Throat Area: %.6f m^2\\n', best_design.At);
166     fprintf('Nozzle Exit Area: %.4f m^2\\n', best_design.Ae);
167     fprintf('Wedge1 Length: %.4f m\\n', best_design.L_wedge1);
168     fprintf('Wedge2 Length: %.4f m\\n', best_design.L_wedge2);
169
170     fprintf('\\nStation Properties (M, P [kPa], T [K], Pt [kPa], Tt [K]):\\n');
171     stations = {'0 (Free)', '1 (Obli1)', '2 (Obli2)', '3 (Norm)', '4 (Diff)', '5 (
        Comb)', '6 (Throat)', '7 (Exit)'};
172     for i = 1:8

```

```

173     fprintf('%-10s: M=%.2f, P=%7.2f, T=%6.1f, Pt=%7.2f, Tt=%6.1f\n', ...
174         stations{i}, best_design.M(i), best_design.P(i)/1000, best_design.T(i),
175         ...
176         best_design.Pt(i)/1000, best_design.Tt(i));
177 end
178
179 fprintf('\n===== \n');
180 fprintf('BONUS: OFF-DESIGN (Fixed Geom) \n');
181 fprintf('===== \n');
182 theta1_fix = best_design.theta1;
183 theta2_fix = best_design.theta2;
184 At_fix = best_design.At;
185 Ae_fix = best_design.Ae;
186
187 off_design_Machs = [2.0, 4.0];
188
189 for M_off = off_design_Machs
190     fprintf('\n--- Analyzing Off-Design Mach %.1f --- \n', M_off);
191
192     Tt0_off = T_inf * (1 + (gamma-1)/2 * M_off^2);
193     Pt0_off = P_inf * (1 + (gamma-1)/2 * M_off^2)^(gamma/(gamma-1));
194
195     V_off = M_off * sqrt(gamma*R*T_inf);
196     mdot_off = (P_inf/(R*T_inf)) * V_off * A_cross_max;
197
198     [~, M1_o, pt1_r, v1] = solve_oblique_robust(M_off, theta1_fix, gamma);
199
200     if v1
201         [~, M2_o, pt2_r, v2] = solve_oblique_robust(M1_o, theta2_fix, gamma);
202         if v2
203             [~, pt3_r] = solve_normal(M2_o, gamma);
204             Pt3_off = Pt0_off * pt1_r * pt2_r * pt3_r;
205             Tt4_off = Tt0_off;
206
207             T_sweep = linspace(Tt4_off, T_comb_max, 100);
208             max_F_off = -inf;
209             opt_T_off = Tt4_off;
210
211             for T_try = T_sweep
212
213                 M4_est = 0.22;
214                 ray_rat_4 = (2*(gamma+1)*M4_est^2 * (1 + (gamma-1)/2*M4_est^2)) /
215                     (1 + gamma*M4_est^2)^2;
216                 Tt_star = Tt4_off / ray_rat_4;
217
218                 if T_try > Tt_star, T_actual = Tt_star; else, T_actual = T_try;
219                 end
220
221                 ray_rat_5 = T_actual / Tt_star;
222                 M5_off = solve_rayleigh_M(ray_rat_5, gamma);

```

```

222     pt_rat_4 = (gamma+1)/(1+gamma*M4_est^2) * ((2/(gamma+1)) * (1+(
223         gamma-1)/2*M4_est^2))^(gamma/(gamma-1));
224     pt_rat_5 = (gamma+1)/(1+gamma*M5_off^2) * ((2/(gamma+1)) * (1+(
225         gamma-1)/2*M5_off^2))^(gamma/(gamma-1));
226
227     Pt5_off = 0.95 * Pt3_off * (pt_rat_5 / pt_rat_4);
228
229     Ae_At = Ae_fix / At_fix;
230     [Me_off, ~] = solve_area_mach(Ae_At, gamma, 'supersonic');
231     Pe_off = Pt5_off * (1 + (gamma-1)/2*Me_off^2)^(-gamma/(gamma-1));
232     Ve_off = Me_off * sqrt(gamma * R * (T_actual / (1 + (gamma-1)/2*
233         Me_off^2)));
234
235     F_net_now = mdot_off * (Ve_off - V_off) + (Pe_off - P_inf) *
236         Ae_fix;
237
238     if F_net_now > max_F_off
239         max_F_off = F_net_now;
240         opt_T_off = T_actual;
241     end
242     fprintf('Optimal_Combustor_Temp: %.1f_K\n', opt_T_off);
243     fprintf('Max_Net_Thrust: %.2f_N\n', max_F_off);
244 else
245     fprintf('Intake_Unstart_(Shock_2_detached)\n');
246 end
247 else
248     fprintf('Intake_Unstart_(Shock_1_detached)\n');
249 end
250 end
251
252 function [beta, M2, Pt_ratio, valid] = solve_oblique_robust(M1, theta_deg, gamma)
253 valid = false; beta=0; M2=0; Pt_ratio=0;
254
255 if M1 <= 1 || theta_deg <= 0, return; end
256 theta = deg2rad(theta_deg);
257 mu = asin(1/M1);
258 beta_grid = linspace(mu+1e-4, pi/2-1e-4, 500);
259 theta_grid = atan( 2*cot(beta_grid) .* (M1^2 .* sin(beta_grid).^2 - 1) ./ ...
260     (M1^2 .* (gamma + cos(2*beta_grid)) + 2) );
261 if theta > max(theta_grid), return; end
262 f = theta_grid - theta;
263 idx = find(f(1:end-1).*f(2:end) <= 0, 1, 'first');
264 if isempty(idx), return; end
265 try
266     beta = fzero(@(b) atan( 2*cot(b) * (M1^2*sin(b)^2 - 1) / (M1^2*(gamma + cos(2*b))
267         + 2) ) - theta, ...
268         [beta_grid(idx), beta_grid(idx+1)]);
269 catch
270     return;
271 end
272 Mn1 = M1*sin(beta);
273 if Mn1 <= 1, return; end

```



```

270 Mn2 = sqrt( (1 + (gamma-1)/2 * Mn1^2) / (gamma*Mn1^2 - (gamma-1)/2) );
271 M2 = Mn2 / sin(beta - theta);
272 term1 = ((gamma+1)*Mn1^2 / (2 + (gamma-1)*Mn1^2))^(gamma/(gamma-1));
273 term2 = (gamma+1) / (2*gamma*Mn1^2 - (gamma-1));
274 Pt_ratio = term1 * term2^(1/(gamma-1));
275 valid = true;
276 end
277 function [M2, Pt_ratio] = solve_normal(M1, gamma)
278 M2 = sqrt( (1 + (gamma-1)/2 * M1^2) / (gamma*M1^2 - (gamma-1)/2) );
279 term1 = ((gamma+1)*M1^2 / (2 + (gamma-1)*M1^2))^(gamma/(gamma-1));
280 term2 = (gamma+1) / (2*gamma*M1^2 - (gamma-1));
281 Pt_ratio = term1 * term2^(1/(gamma-1));
282 end
283 function M = solve_rayleigh_M(ratio, gamma)
284 fun = @(m) ratio - (2*(gamma+1)*m^2 * (1 + (gamma-1)/2*m^2)) / (1 + gamma*m^2)^2;
285 try, M = fzero(fun, [0.001, 0.999]); catch, M = 0.999; end
286 end
287 function [M, valid] = solve_area_mach(AR, gamma, branch)
288 fun = @(m) AR - (1/m) * ((2/(gamma+1)) * (1 + (gamma-1)/2*m^2))^((gamma+1)/(2*(gamma-1)));
289 valid = true;
290 try
291     if strcmp(branch, 'supersonic')
292         M = fzero(fun, [1.001, 20]);
293     else
294         M = fzero(fun, [0.001, 0.999]);
295     end
296 catch
297     M = 1; valid = false;
298 end
299 end

```